COURSE

## Pearson Edexcel Level 3 Advanced GCE in Business (9BS0)

# Investment appraisal

**Theme 3: Business decisions and strategy · 3.3 Decision-making techniques · 3.3.2 Investment appraisal**

Revision for Theme 3: Business decisions and strategy: each firm works it out on its own deals.

| Class | Topic |
|---|---|
| **Pearson Edexcel A level Business (9BS0)** | **Investment appraisal** |

THE LESSON

**Setup** — 3 min

**Round 1** — Decisions 3 min · Deals 5 min · Topic tasks 4 min

| Firm decisions | Deals | TASK 1 Average profit | TASK 2 Limits of payback |
|---|---|---|---|

**Round 2** — Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 3 min

| Firm decisions | Deals | TASK 3 Rank by payback | TASK 4 Payback in months | B2B deal |
|---|---|---|---|---|

**Round 3** — Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 3 min

| Firm decisions | Deals | TASK 5 Ramp or bay | TASK 6 Quicker payback | B2B deal |
|---|---|---|---|---|

**Final questions** — 3 min

Fits a 30-minute lesson setting; longer settings add more deals.

# Each step

| | |
|---|---|
| **Firm decisions** | Each role makes its decision on its own page. |
| **Deals** | Each pupil trades with a classmate in another firm. |
| **Topic task** | The firm answers together. Everyone presses Agree. |
| **B2B deal** | One firm sells a lot of cars to another firm. |
| **Final questions** | Each pupil answers six questions. |

# Firm decisions

| | | |
|---|---|---|
| **CEO** | **Staff** | Hire staff. Each costs a salary and gives one benefit. |
| | **Sabotage** FROM ROUND 2 | Hit a rival firm's cash, at a cost. A Legal Counsel blocks it. |
| **Sales & Marketing** | **Marketing** | Book one advert campaign to lift the next sales. |
| **Finance** | **Capital & Loans** | Borrow cash. The firm pays it back with interest. |
| | **Investments** FROM ROUND 2 | Buy property or funds. The value can rise or fall. |
| **Operations** | **Seasonal stock** | Order the season's cars in advance. |

**B2B deal · Rounds 2 and 3**

One firm sells a lot of 3 cars to another firm (4 in Round 3). With an odd number of firms, two firms offer to sell and the lower price wins.

# Topic tasks

Each firm gets its own figures. The method is the same for every firm.

## Task 1

ROUND 1

New charging points.

**Average profit each year?**

WORK OUT THE NUMBER · EXAMPLE FIGURES

| | |
|---|---|
| Cost to set up | **£30,000** |
| Returns over its life | **£66,000** |
| It lasts | **4 years** |

METHOD

**Average profit = (returns – set-up cost) ÷ years**

(£66,000 – £30,000) ÷ 4 = £9,000

**Why:** Average yearly profit is total profit shared over the years.

## Task 2

ROUND 1

You choose between two projects by payback.

**Which two are limits of payback?**

PICK TWO

- ○ It favours slow projects
- ✔ **It ignores returns after payback**
- ○ It ignores the project's cost
- ✔ **It ignores the time value of money**
- ○ It is hard to work out

**Why:** Payback stops at the payback point and treats a pound later as a pound now.

## Task 3

ROUND 2

Three projects with even yearly returns.

**Order by payback, fastest first?**

PUT THESE IN ORDER, FIRST TO LAST · RIGHT ORDER

1. **Cost £30,000, £12,000 a year**
2. **Cost £60,000, £20,000 a year**
3. **Cost £40,000, £10,000 a year**

**Why:** Payback: 2.5, 3 and 4 years.

## Task 4

ROUND 2

A new valeting bay.

**Payback period, in months?**

WORK OUT THE NUMBER, THEN PICK IT · EXAMPLE FIGURES

| | |
|---|---|
| Cost to set up | **£17,100** |
| Returns, each year | **£8,290** |
| Running costs, each year | **£1,450** |

METHOD

**Payback = cost ÷ (returns – running costs) × 12 months**

£8,290 − £1,450 = £6,840; £17,100 ÷ £6,840 × 12 = 30

**Why:** It has paid for itself after 30 months.

## Task 5

ROUND 3

Ramp: payback 2 years, ARR 10%. Bay: payback 4 years, ARR 18%. Demand is stable; cash is fine.

**Which project should you choose, and why?**

PICK A OR B AND GIVE A REASON

- ✔ **The bay**
- ○ The ramp

**Why:** With little risk and cash to spare, the higher return wins.

## Task 6

ROUND 3

Which pays back first?

**Payback of the quicker option, in months?**

WORK OUT THE NUMBER · EXAMPLE FIGURES

| | |
|---|---|
| Option A: cost | **£17,700** |
| Option A: returns a year | **£7,080** |
| Option B: cost | **£38,880** |
| Option B: returns a year | **£12,960** |

METHOD

**Each option: cost ÷ returns × 12 months; take the quicker**

A 30 months; B 36 months; quicker 30

**Why:** The shorter payback gets the money back sooner.

# Final questions

**6**
questions for each pupil

**109**
Level 3 Business questions

**7**
of them on investment appraisal

**The game draws 6 of the 109 Level 3 questions; these 7 are about investment appraisal.**

1. An inspection line costs £300,000 and should make £45,000 average annual profit. On original cost, what is its accounting rate of return?
   **15%.**

2. A £180,000 paint booth returns £60,000, £75,000 and £90,000 over three years. Ignoring timing within years, what is its payback period?
   **2.5 years.**

3. A project requires £120,000 now and has discounted future cash inflows of £138,500. What is its net present value?
   **£18,500.**

4. A project costs £200,000 and its discounted future cash inflows total £224,000. What is NPV?
   **£24,000.**

5. A system costs £250,000 and is expected to earn average annual profit of £30,000. What is ARR using original cost?
   **12%.**

6. An energy-saving project costs £180,000. Discounted savings are £205,000, but a downside scenario reduces those savings by 15%. What is NPV in the downside scenario?
   **−£5,750.**

7. A £90,000 paint booth brings in £30,000 in year 1, then £40,000 a year. What is its payback period?
   **2.5 years.**